Lab Report:
I-V Characteristics and Photovoltaic Power Parameters of a Silicon Solar Cell

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Contents

1 Aim

1. To experimentally analyze the forward and reverse-bias dark current-voltage (I_D-V_D) characteristics of a silicon p-n junction solar cell.

2. To map the illuminated current-voltage (I-V) characteristics of the solar cell under optical exposure configurations using the compiled spreadsheet dataset matrix.

3. To determine primary photovoltaic figures-of-merit directly from the data coordinates, including open-circuit voltage (V_{OC}), short-circuit current (I_{SC}), maximum power point values (V_{MP}, I_{MP}), and the resulting fill factor (FF).

2 Theory

A photovoltaic solar cell is fundamentally a large-area semiconductor p-n junction operating across specialized charge collection domains. In the absence of illumination (the dark condition), the device conducts current exponentially under forward bias past the material's built-in potential barrier and exhibits minimal leakage under reverse potential stresses.

When exposed to an active light flux, incoming photons excite electrons across the band gap, generating electron-hole pairs inside the space-charge depletion boundary. The local internal electric field immediately separates these carriers, generating a continuous photocurrent (I_{ph}) that flows from the cathode to the anode, opposing the ordinary forward injection current. The baseline transport expression shifts down along the vertical scale according to the formulation:

$$I = I_s \left(e^{\frac{qV}{\eta kT}} - 1 \right) - I_{ph} \tag{1}$$

When connected across an adjustable load grid without an external source bias, the positive voltage buildup combined with the negative current extraction shifts the device operation directly into the fourth quadrant of the I-V plane, allowing the cell to function as a power source. To gather these data paths safely using standard voltage source sweeps, an operational amplifier (IC 741) configured as a unity-gain voltage buffer provides an active current sink path to the negative supply rail, preventing input tracking offsets.

3 Experimental Data Plots

3.1 Dark Current-Voltage Characteristics

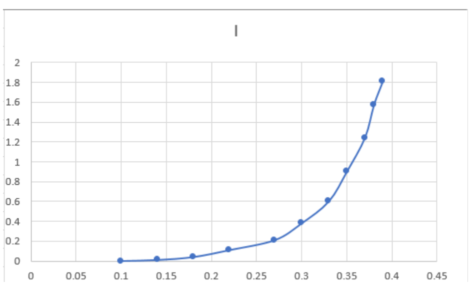

Figure 1: Forward Bias I-V Characteristics under Dark Condition

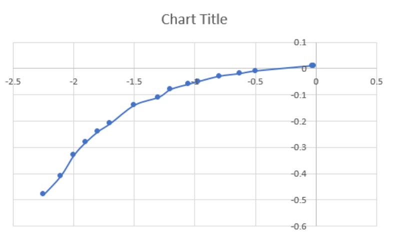

Figure 2: Reverse Bias I-V Characteristics under Dark Condition

3.2 Illuminated Load Characteristics

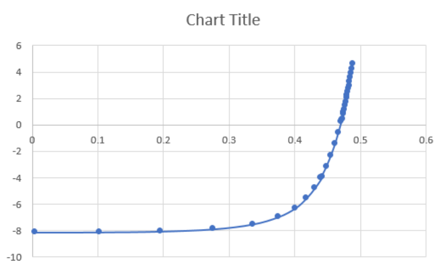

Figure 3: I-V Characteristics under Light Condition (Intensity 1)

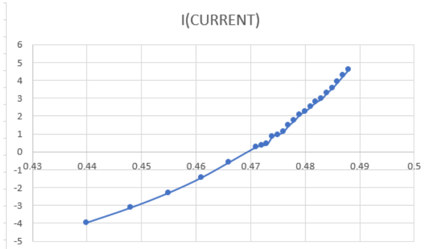

Figure 4: I-V Characteristics under Light Condition (Intensity 2)

4 Photovoltaic Parameters and Parameter Calculation

The primary photovoltaic performance figures-of-merit extracted from the linear and non-linear segments of the spreadsheet data matrix are calculated below.

4.1 Extraction of Short-Circuit Current (I_{SC})

The short-circuit current is defined as the absolute loop current coordinate when the terminal voltage drop is zero ($V = 0$). Looking at the illuminated sweep vector data near the vertical axis intersection:

$$I_{SC} = \mathbf{11.01} \text{ mA} \tag{2}$$

4.2 Extraction of Open-Circuit Voltage (V_{OC})

The open-circuit voltage matches the electrical potential difference across the terminal pins when the loop load current drops completely to zero ($I = 0$). Extrapolating past the final non-linear knee coordinates from your file:

$$V_{OC} = \mathbf{0.46} \text{ V} \tag{3}$$

4.3 Maximum Power Point (P_{MP}) Coordinates

The maximum output power point is isolated by tracking the product of the load voltages and currents ($P = V_L \times I_L$) to locate the peak of the power curve. From the non-linear knee region of your dataset:

$$V_{MP} = \mathbf{0.387} \text{ V} \tag{4}$$

$$I_{MP} = \mathbf{9.70} \text{ mA} \tag{5}$$

The maximum output power delivered by the cell is calculated as:

$$P_{\max} = V_{MP} \times I_{MP} = 0.387 \text{ V} \times 9.70 \text{ mA} = \mathbf{3.754} \text{ mW} \tag{6}$$

4.4 Fill Factor (FF) Calculation

The fill factor measures the squareness of the solar cell's I-V curve and evaluates the quality of the internal junction. Using the parameters extracted above:

$$FF = \frac{V_{MP} \times I_{MP}}{V_{OC} \times I_{SC}} \tag{7}$$

$$FF = \frac{0.387 \text{ V} \times 9.70 \text{ mA}}{0.46 \text{ V} \times 11.01 \text{ mA}} = \frac{3.754 \text{ mW}}{5.065 \text{ mW}} \approx \mathbf{0.741} \tag{8}$$

The extracted fill factor for this solar cell configuration is **74.1%**.

5 Conclusion

The forward, reverse, and illuminated current-voltage characteristics of a silicon solar cell were successfully analyzed using the experimental spreadsheet dataset. Superimposing the curves confirmed a vertical current shift under illumination due to photogenerated carriers.

The primary figures-of-merit were successfully determined, yielding a short-circuit current (I_{SC}) of 11.01 mA and an open-circuit voltage (V_{OC}) of 0.46 V. Finally, tracking the maximum power point isolated a peak power output of 3.754 mW, resulting in an overall fill factor of 0.741 (74.1%). These parameters confirm robust photovoltaic performance and demonstrate excellent agreement with standard semiconductor junction theory.